

Member ID: _____

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LINUX OPERATING SYSTEM FUNDAMENTALS

(350)

NATIONAL 2023

Total Points: (50 @ 2 points each) _____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. Which Linux command would allow you to check how long the system has been running?
 - A. uptime
 - B. top
 - C. ps
 - D. ontime
2. When using the FirewallD firewall what command below would open up port 22?
 - A. firewall-cmd --add-port=22/tcp
 - B. firewall-cmd --open-port=22/tcp
 - C. firewall-cmd --enable-port=22/tcp
 - D. firewall-cmd --allow-port=22/tcp
3. Which Linux command line tools can be used to copy files securely over SSH?
 - A. ftp
 - B. scp
 - C. rsync
 - D. Both B and C
4. How can you copy a folder named "Documents" and its contents to another system named "ServerB" using scp?
 - A. scp -r Documents ServerB:~
 - B. scp Documents ServerB:~
 - C. scp -R Documents ServerB:~
 - D. scp -cp Documents ServerB:~
5. What command can you use to view a list of all users on a RedHat Linux system?
 - A. userlist
 - B. who
 - C. cat /etc/passwd
 - D. ls /home/
6. How can you block NFS traffic using firewall in Redhat Linux?
 - A. firewall-cmd --add-service=nfs
 - B. firewall-cmd --block-service=nfs
 - C. firewall-cmd --remove-service=nfs
 - D. firewall-cmd --disable-service=nfs

7. A server called **ServerA** is exporting an NFS share from the **/Exports** directory, and you need to mount it to the local **/mnt/data** folder on your Redhat Linux workstation. Would the command seen below correctly mount the NFS share?
mount -t nfs ServerA:/Exports /mnt/data
 - A. Yes
 - B. No
8. Which command below would correctly mount unprotected samba share from **ServerA** called **UserData** to the local directory **/mnt/users**.
 - A. mount -t smb //ServerA/UserData /mnt/users
 - B. mount -t smb ServerA:/UserData /mnt/users
 - C. mount -t cifs //ServerA/UserData /mnt/users
 - D. mount -t cifs ServerA:/UserData /mnt/users
9. Where in Linux do you configure NFS exports?
 - A. /etc/nfs
 - B. /etc/exports
 - C. /etc/share
 - D. /etc/config/nfs
10. Which entry below could you add to your fstab file to correctly mount the NFS share **/Exports** on **ServerA** to the local directory **/mnt/data** at boot?
 - A. ServerA:/Exports /mnt/data nfs
 - B. /mnt/data ServerA:/Exports nfs
 - C. //ServerA/Backup /mnt/data nfs
 - D. /mnt/data /Exports:ServerA nfs
11. What is the utility used to manage or edit cron jobs in Linux?
 - A. cron
 - B. cronjob
 - C. crontab
 - D. cronedit
12. Which file would you edit to add provide cronjob privileges to a specific user?
 - A. /etc/crontab
 - B. /etc/cron.allow
 - C. /etc/cronuser
 - D. /etc/cron.access
13. Where would you place a program or script that you wanted to have the cron service run every hour?
 - A. /etc/cron.d/hourly
 - B. /etc/cron.hourly
 - C. /etc/cron.d/crontab
 - D. /etc/cronjob

14. Which below could you use to check the link speed of a network interface named eth0?
 - A. ethtool eth0
 - B. ip link
 - C. ethtool -s eth0
 - D. ifconfig
15. How can you view the currently loaded modules in the Linux kernel?
 - A. showmod
 - B. modprobe
 - C. lsmod
 - D. listmod
16. In Linux the utility mdadm can be used to manage software RAID devices.
 - A. True
 - B. False
17. How can you check the dependencies for the package openssh-server before attempting to install on a Redhat Linux system?
 - A. apt depends openssh-server
 - B. apt list openssh-server
 - C. yum deplist openssh-server
 - D. yum list openssh-server
18. Which Linux command can you use to list all NFS version 4 exports shared on the current system?
 - A. exportfs
 - B. showmount
 - C. shownfs
 - D. nfslist
19. Which command can you use to delete an LVM logical volume?
 - A. delvol
 - B. lvdelete
 - C. lvremove
 - D. rmvol
20. What command below would allow you to remove a member disk, **sdb1**, from a volume group named **VD1**?
 - A. vgreduce /dev/sdb1 VD1
 - B. vgremove VD1 /dev/sdb1
 - C. vgremove /dev/sdb1 VD1
 - D. vgreduce VD1 /dev/sdb1
21. Which of these Linux filesystems does **NOT** support shrinking?
 - A. ext4
 - B. btrfs
 - C. xfs
 - D. reiserfs

22. Which command below would increase a logical volume named data on a volume group named vg0 by adding an additional 5Gb to it?

- A. lvextend vg0/data -L 5G
- B. lvextend vg0/data -L +5G
- C. lvresize -l 5G vg0/data
- D. lvresize -l +5G vg0/data

23. Looking at the below output what does the label Shared represent?

Data Shared -wi-ao---- 25.00t

- A. Logical Volume
- B. Volume Group
- C. Physical Volume
- D. RAID volume

24. The output listed below is seen from running which command?

```
/dev/mapper/rhel-root: UUID="921153c0-889c-4470-be1a-c5b6f369f7be" BLOCK_SIZE="512" TYPE="xfs"
/dev/sdb2: UUID="MnZAX0-nCcn-6j3E-t5jJ-8fRq-jonz-IvfD52" TYPE="LVM2_member" PARTUUID="d85745a3-02"
/dev/sdb1: UUID="d786f076-f465-4881-b333-6505fa5a572c" BLOCK_SIZE="512" TYPE="xfs" PARTUUID="d85745a3-01"
/dev/mapper/rhel-swap: UUID="a556f207-bd8f-4d95-9e35-1a3571a9ff3d" TYPE="swap"
```

- A. lsblk
- B. lvs
- C. blkid
- D. pvs

25. Which command will show the health of a Linux software RAID disk labeled as “md1”?

- A. mdadm -M /dev/md1
- B. mdadm -H /dev/md1
- C. mdadm -D /dev/md1
- D. mdadm -L /dev/md1

26. The output seen below was provided by running the command lvs.

```
NAME      MAJ:MIN RM   SIZE RO TYPE  MOUNTPOINT
sda        8:0    0    24G  0 disk
└─sda1     8:1    0    24G  0 part /
sdb        8:16    0    16G  0 disk
└─md0      9:0    0    16G  0 raid1
sdc        8:32    0    16G  0 disk
└─md0      9:0    0    16G  0 raid1
sr0       11:0    1 1024M  0 rom
```

- A. True
- B. False

27. When trying to remotely login to a Linux server named ServerA via ssh you see the response shown below. Which would explain why you received this response?

ssh: connect to host ServerA port 22: Connection refused

- A. SSH service is not running on ServerA
 - B. Firewall on ServerA blocking port 22
 - C. The name ServerA is not known by DNS
 - D. All of the above
28. When trying to mount an NFS share “/Export” from ServerA you received the error seen below. Which would be a valid reason for seeing this error?

mount.nfs: access denied by server while mounting ServerA:/Export

- A. No entry in NFS exports file for /Export
 - B. The NFS service is not running on ServerA
 - C. The firewall on ServerA is blocking NFS
 - D. Your system is not listed in ServerA’s permissions for /Export
29. The admin at your company has set folder permissions for multiple user groups, which command can you use to view the folder permissions for each group?

- A. ls
- B. stat
- C. getfacl
- D. showfacl

30. You need to set permissions for two different users to have read/write permissions to the same file and the two users are not in the same group. Which command below would allow you to do this?

- A. chown
- B. chmod
- C. setfacl
- D. usermod

31. Which option would you add with the command setfacl to change the set ACLs on a folder?

- A. -f
- B. -d
- C. -m
- D. -s

```
# file: Marketing/
# owner: root
# group: root
user::rwx
user:tom:---
user:larry:r-x
user:susan:rwx
group:---
group:admins:rwx
mask::rwx
other:---
```

Image 3A: In the Image above you can see the ACL permissions for the folder Marketing on our Linux system. The group admins contain both users Tom and Larry.

32. Referring to the ACL's seen in **Image 3A** above which user(s) would **NOT** have access to view files in the Marketing folder?
 - A. Tom
 - B. Susan
 - C. Larry
 - D. None of the above
33. Referring to the above **Image 3A** again which user(s) would **NOT** have permission to add or change files in the Marketing folder?
 - A. Susan
 - B. Larry
 - C. Tom
 - D. Both B and C
34. Which command would give the group Marketing **read** and **execute** permissions to all current and newly created files/folders in the Marketing folder?
 - A. setfacl -m u:Marketing:rx Marketing
 - B. setfacl -m d:g:Marketing:rx Marketing
 - C. setfacl -a d:g:Marketing:rx Marketing
 - D. setfacl -a g:Marketing:rx Marketing
35. The **-b** option when used with the Linux command setfacl would remove ALL user/group permissions on a file/folder.
 - A. True
 - B. False
36. Which permissions would a user or group need to move a file out of a folder in Linux?
 - A. Read
 - B. Write
 - C. Execute
 - D. All of the above

37. Which nmcli command below would properly change the Network adapter **em1** from static to dhcp?
- nmcli connection modify em1 ipv4.method dhcp
 - nmcli connection modify em1 ipv4.method auto
 - nmcli device modify em1 ipv4.method dhcp
 - nmcli device modify em1 ipv4.method auto
38. In current versions of Ubuntu what is the new default utility that manages the network adapters configuration?
- NetworkManager
 - Netplan
 - NetMonitor
 - NetConf
39. Which Linux command would show the MAC address of all network adapters?
- ip state
 - ip link
 - ip show
 - ip list
40. Which command below would configure the network adapter **eno1** with IP and subnet of **192.168.5.2/255.255.255.0** in Redhat Linux?
- nmcli connection modify eno1 ipv4.addresses 192.168.5.2/255.255.255.0
 - nmcli connection modify eno1 ipv4.addresses 192.168.5.2/24
 - nmcli device modify eno1 ipv4.addresses 192.168.5.2/255.255.255.0
 - nmcli device modify eno1 ipv4.addresses 192.168.5.2/24
41. In Redhat Linux in what folder would you find the config files to manually setup a network adapter?
- /etc/sysconfig/network-scripts/
 - /etc/network/
 - /etc/sysconfig/network/
 - /etc/network/

```
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: inactive (dead) since Wed 2021-12-29 22:45:51 CST; 1s ago
     Process: 6084 ExecStart=/usr/sbin/sshd -D $SSH_OPTS (code=exited, status=0/SUCCESS)
    Main PID: 6084 (code=exited, status=0/SUCCESS)
```

Image 4A: Showing the status of sshd service

42. Looking at **Image 4A** above you can see the ssh service is currently not running.
- True
 - False

43. Looking at the **Image 4A** above you can see that the ssh service is not set to start automatically at boot.
- A. True
 - B. False
44. Which command can be used to either lock or disable a user account to prevent logins?
- A. usermod
 - B. chage
 - C. passwd
 - D. All of the above
45. Which command below would add a new user named **Mike** and set its group to **Research**?
- A. usermod Mike -g Research
 - B. useradd -G Research Mike
 - C. useradd -G Mike Research
 - D. usermod -g Research Mike
46. Which command would add the existing user named **Sara** to the group **Design** without changing her default group?
- A. usermod -G Design Sara
 - B. usermod -g Design Sara
 - C. usermod -aG Design Sara
 - D. usermod -ag Design Sara
47. Which package manager below is replacing yum as the default package manager in newer Redhat based Linux OS's?
- A. apt
 - B. zypper
 - C. dnf
 - D. None of the above
48. In Redhat Linux which command would update the package repositories without checking for package updates.
- A. yum update
 - B. yum upgrade
 - C. yum updateinfo
 - D. yum makecache
49. In Redhat Linux the commands “yum update” and “yum upgrade” will perform the same function.
- A. True
 - B. False

50. In Redhat Linux which command would search repositories for any packages matching keyword “ssh”?
- A. yum find ssh
 - B. yum repolist ssh
 - C. yum search ssh
 - D. yum list ssh